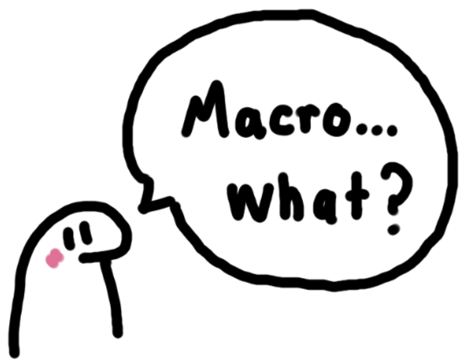
Excel – Macros programming

Introduction

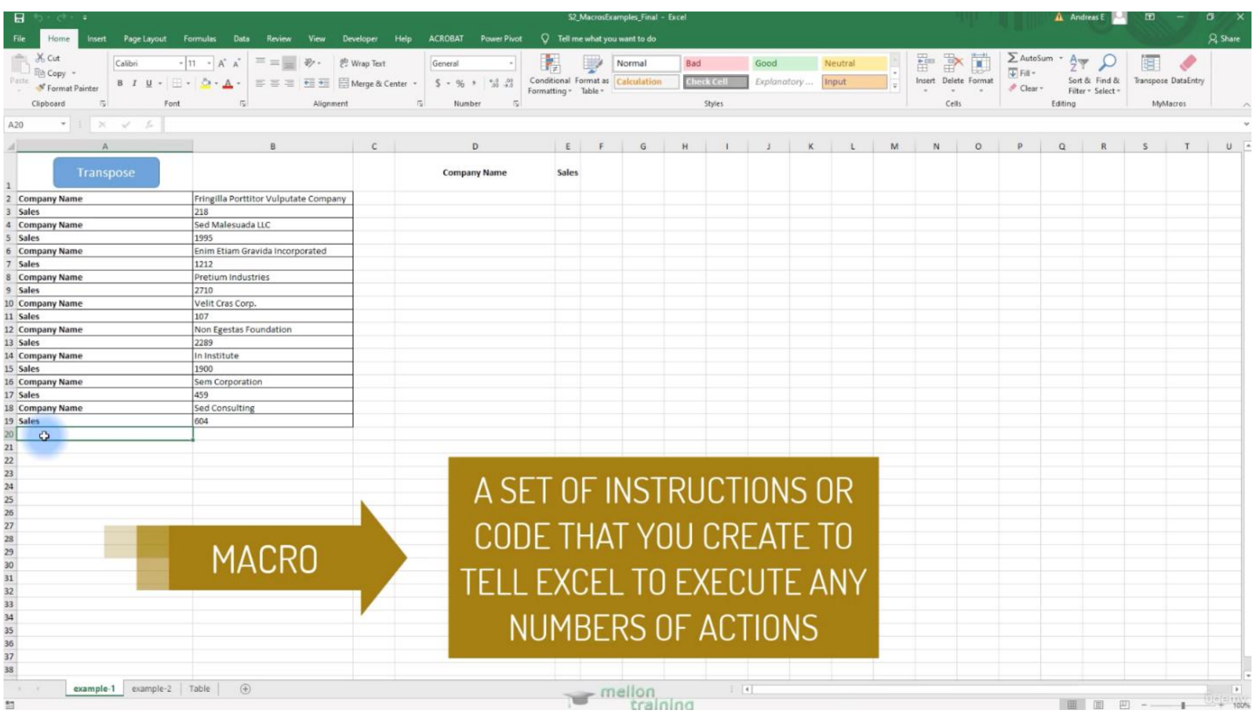
Welcome to the amazing world of Macros in Excel! In this course, you will cover the essential concepts involved in macros programming. Moreover, this course will help you identify and understand the importance of macros in your professional life. Maybe you have heard about macros before, but don't know where to start or how to create them. Well, let's begin!

What are macros?



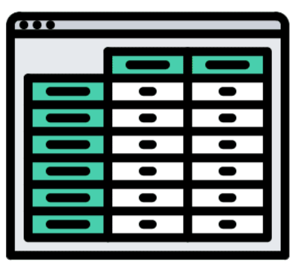
Macros programming in Excel allows you to automate repetitive tasks and perform complex calculations quickly and easily.

A macro is a set of instructions that can be recorded or programmed in Excel, and then it can be executed whenever needed.



It is essentially a series of steps that can be carried out automatically, saving you time and effort.

Macros can be used for a variety of tasks in Excel, such as formatting data, creating charts, or performing calculations.

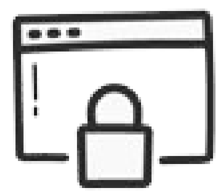


They can save you a lot of time and effort, especially if you need to perform the same task repeatedly.

Macros security – With great power comes great responsibility

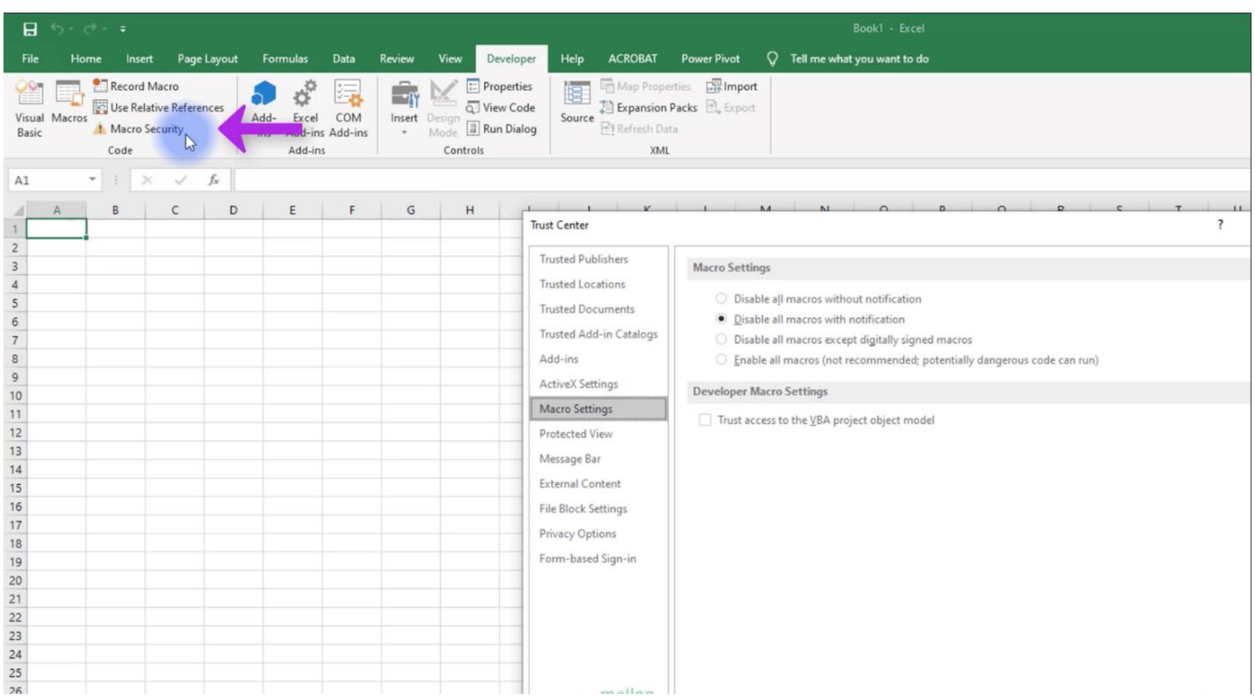
One important thing to keep in mind when working with macros in Excel is to make sure that you trust the source of the macro.



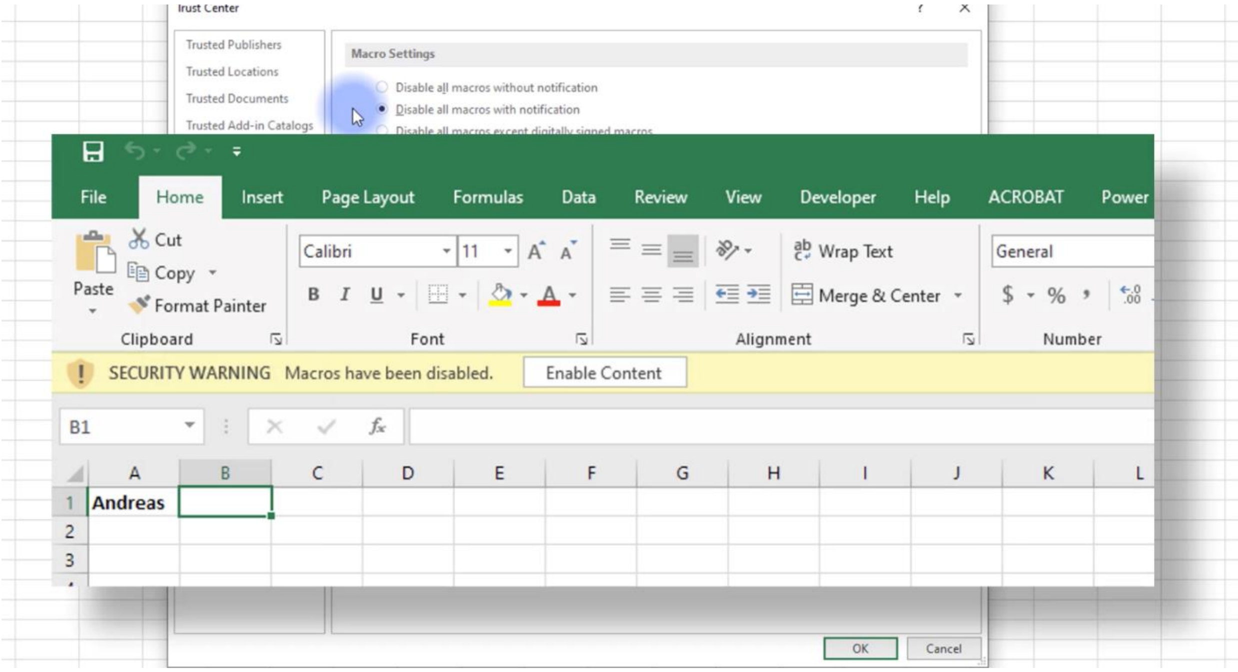


Since macros can execute code on your computer, they can potentially be used to carry out harmful, perform malicious actions, or even do serious damage to your computer.

Therefore, Excel included some macro security features since Excel 2007 to prevent this problem. However, it is important to only use macros from trusted sources and to always enable macro security settings in Excel. To display the macro security settings, choose the developer tab and press the macro security button.



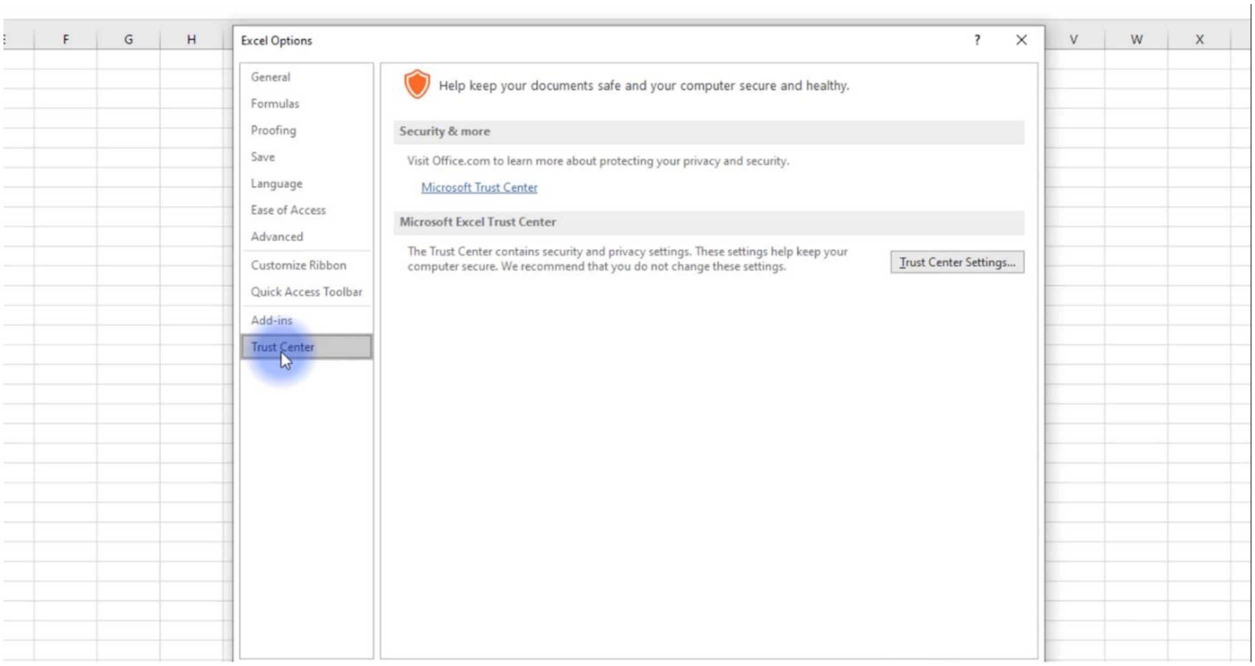
Excel uses by default the 'disable all macros with notification' option. With this setting, if you open a workbook that contains macros, and the file is not digitally signed or stored in a trusted location, Excel displays a warning.

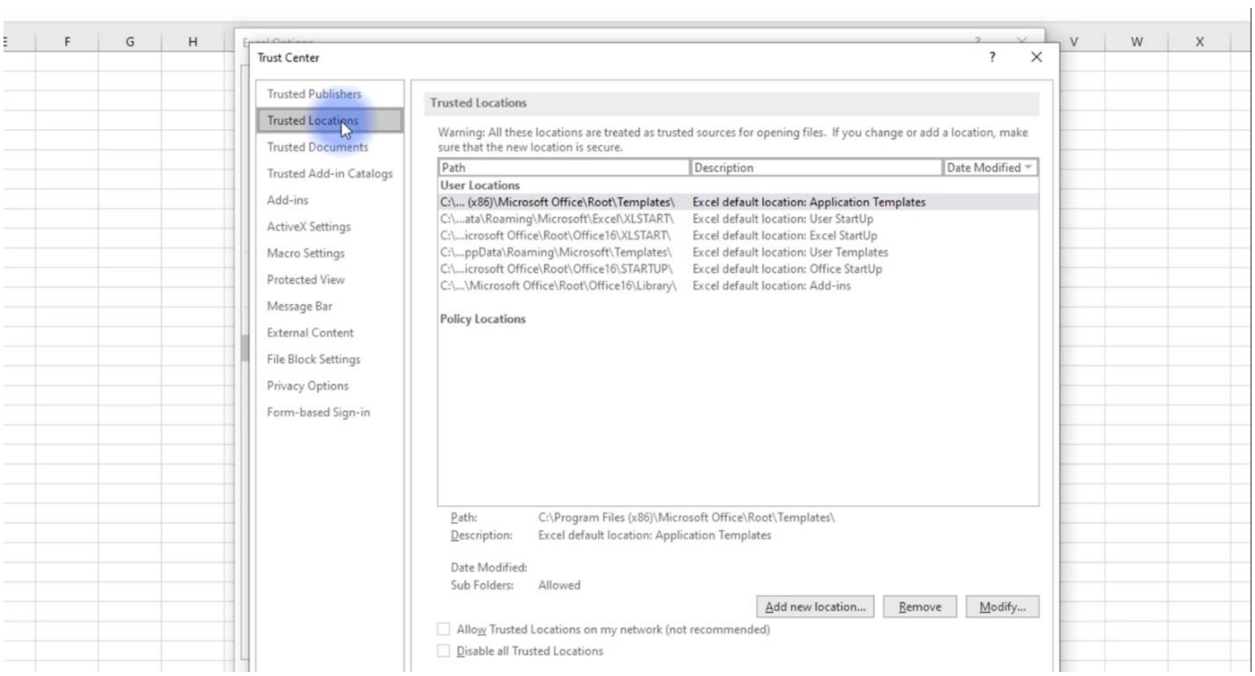


Recommendation:

Perhaps the best way to handle macro security is to designate one or more folders as trusted locations. All the workbooks in the trusted location are open without a macro warning.

You designate trusted folders in the Trusted Locations section of the Trust Center Dialog box.

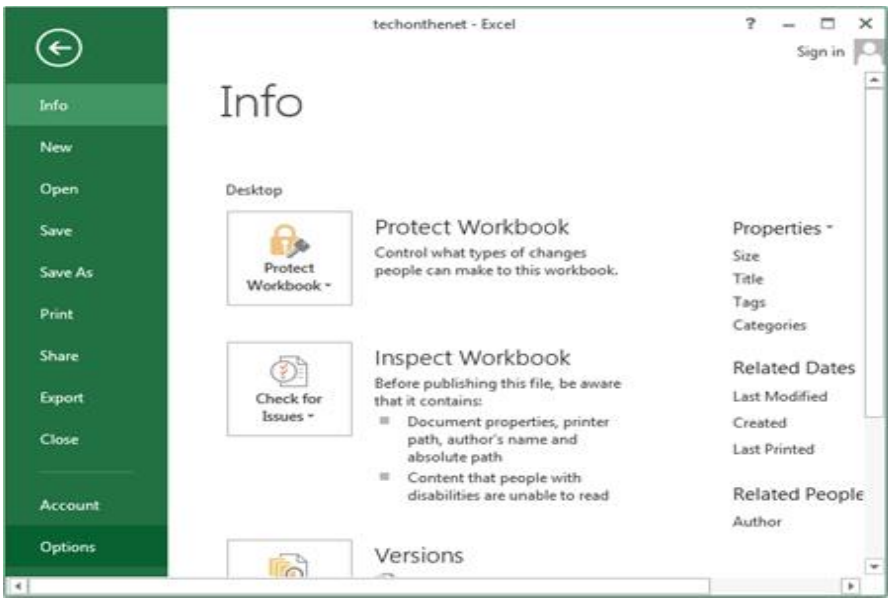


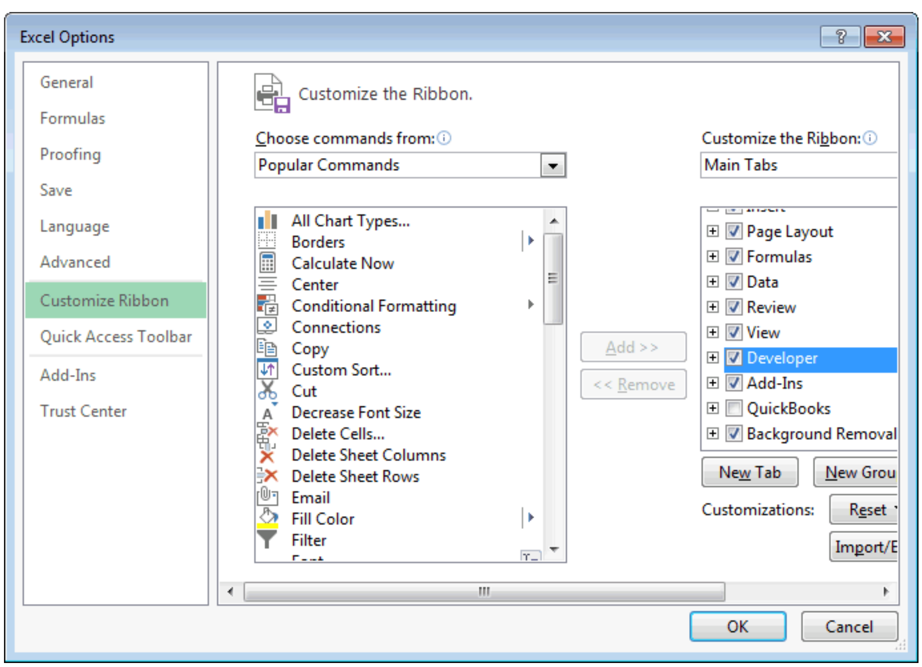


Macro 101 – Release the power in Excel!

To create a macro in Excel, you first need to enable the Developer tab in the Ribbon.

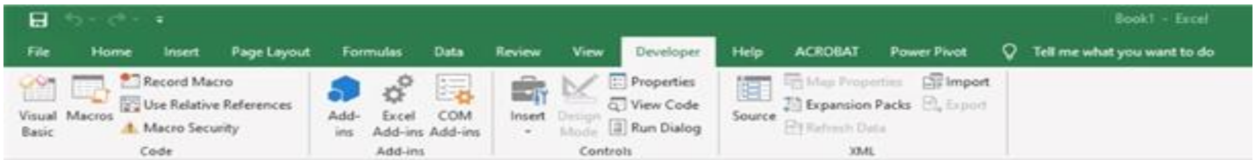
This can be done by going to File > Options > Customize Ribbon, and then checking the box next to Developer.





Once this tab is enabled, you can start recording a macro by going to the Developer tab and clicking on the Record Macro button.

Or you can click on the Visual Basic button.



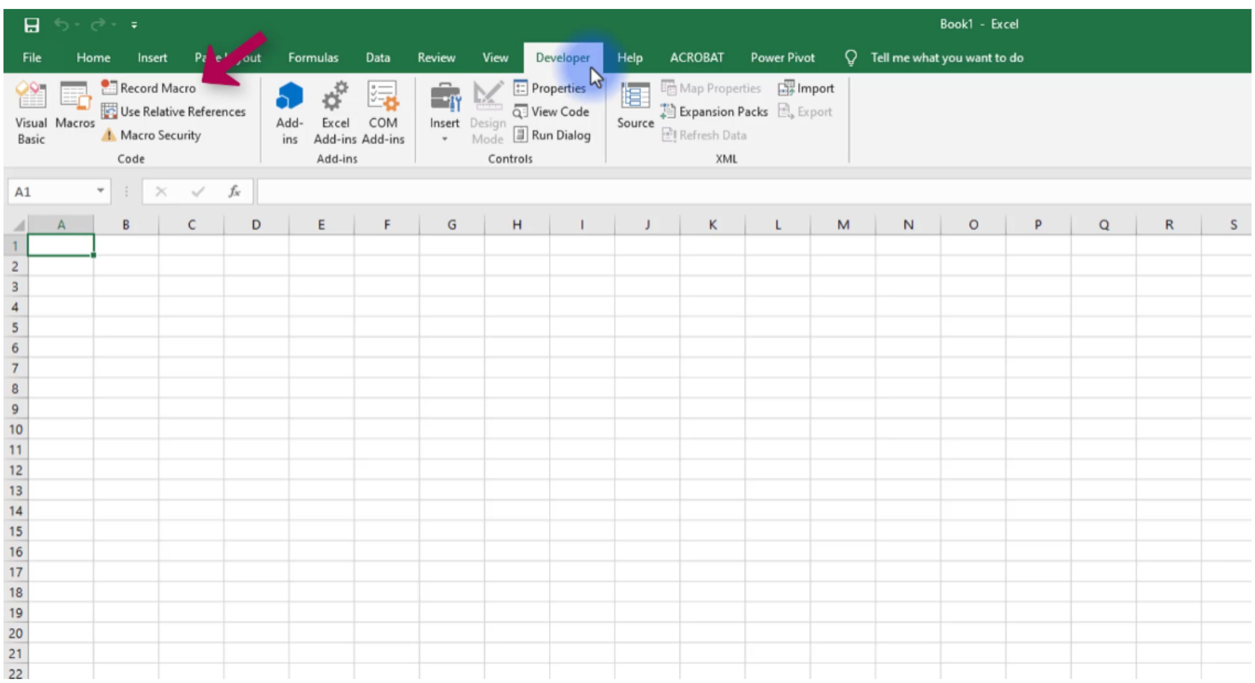
But before we start programming our macros, you must save your file as a special extension of Excel. In other words, your workbook must be saved as an Excel Macro-Enabled Workbook. **This special type of Excel file has a '.xlsm' extension rather than the common '.xlsx' one.**

Macros recording

Macro recording is a feature in software applications that allows users to record a series of steps and then play them back later as a single macro.

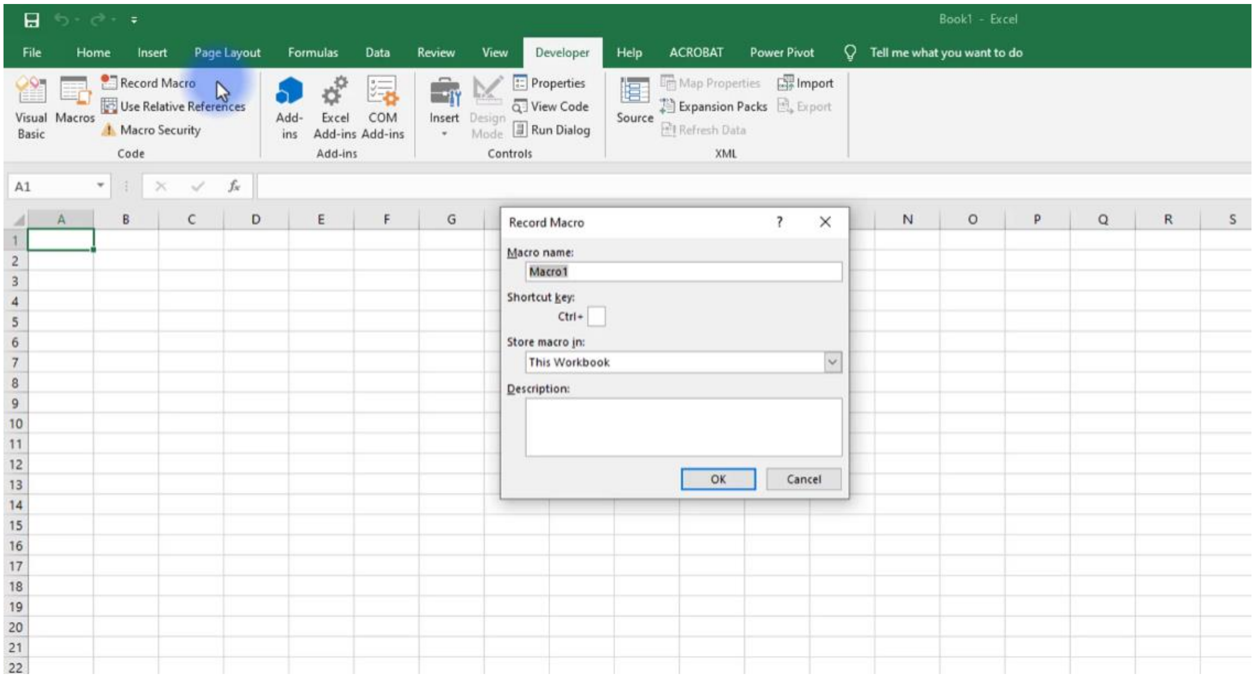
In the context of Microsoft Excel, macro recording allows users to automate repetitive or complex tasks by recording a series of mouse clicks, keyboard strokes, and commands used to perform a particular task.

To start recording a macro, select the ‘Record Macro’ option, and then perform the task or series of tasks you want to automate. Excel will record each action you perform and save them as a macro. Once you are finished, you can stop the recording and save the macro.

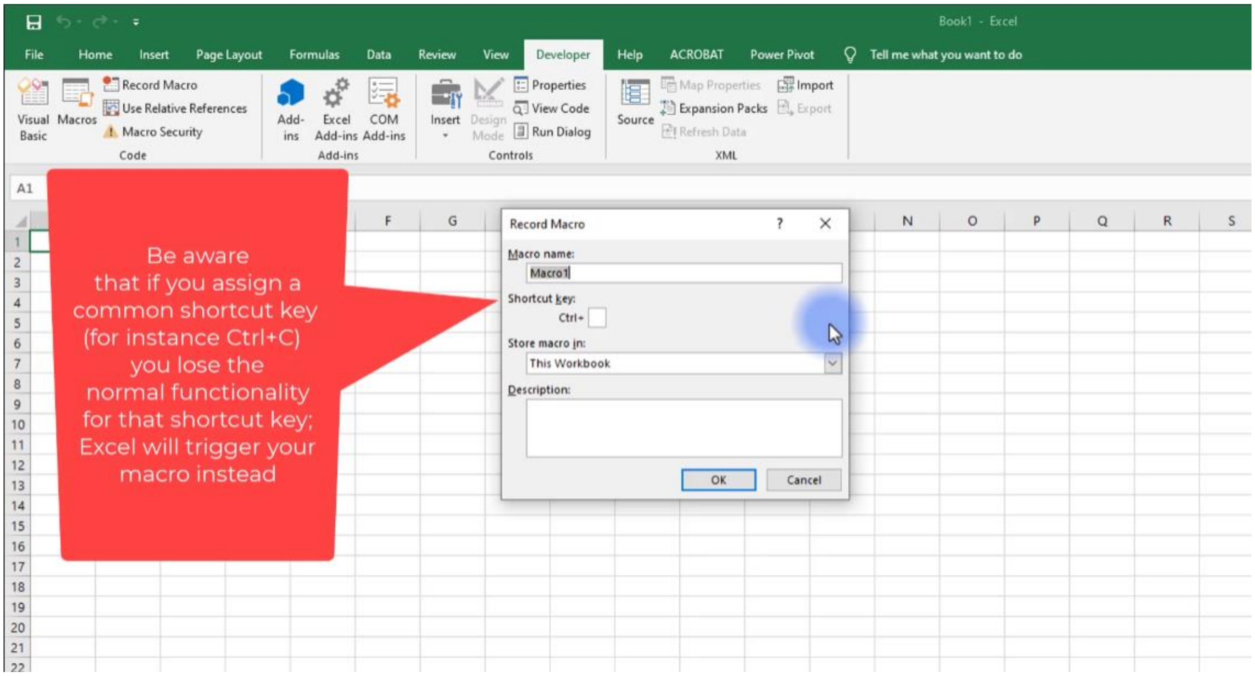


It is worth noting that once you start recording a macro, Excel will start recording EVERYTHING. Therefore, selecting cells, scrolling down the worksheet, etc. will be recorded in the macro.

Excel gives a default name to the macro – Macro1 in this case. A recommendation for naming macros is to try to use descriptive names, as it will be useful to remember what actions the macro performs.



Another thing that you can do before starting to record the macro is that you can assign shortcut keys to execute it later. However, remember that Excel already has certain shortcuts defined.



If you want to use ctrl+C, try instead ctrl+shift+C. This will be interpreted as a capital C, and it will not lose the normal copy function in Excel. At the end of this section, you can find a list of shortcuts that are already defined in Excel. Hope this will be useful to you!

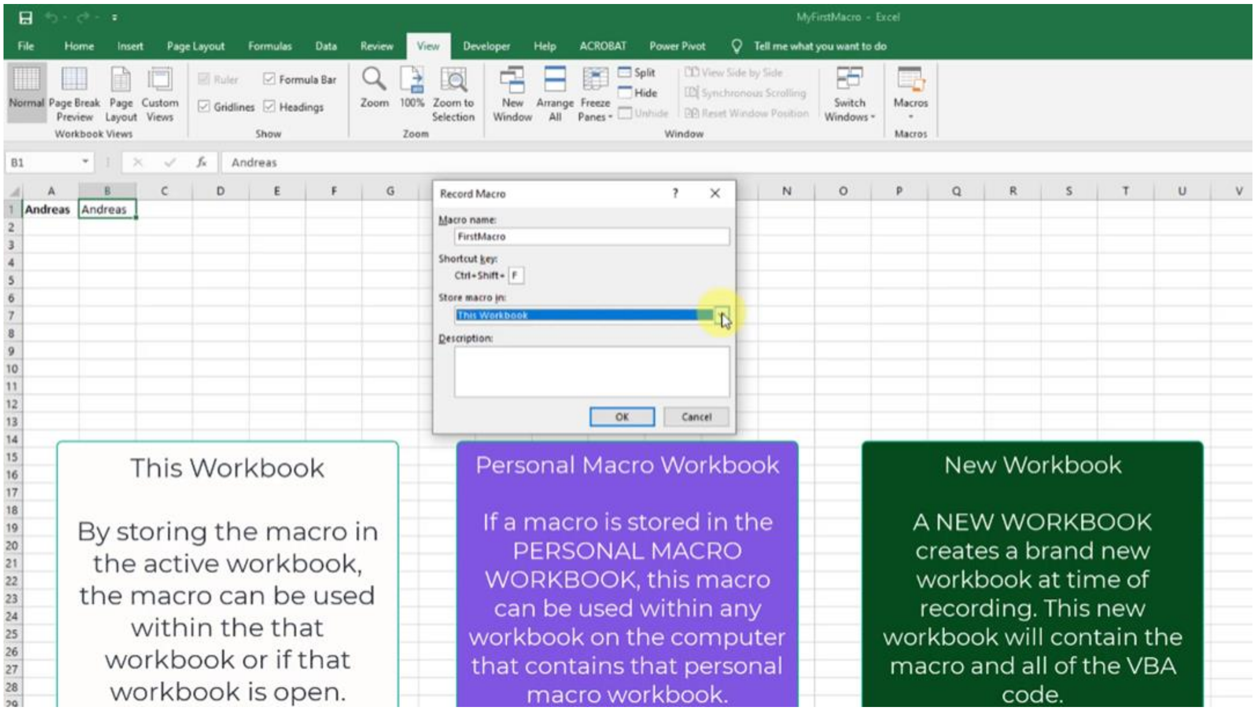
Excel predefined shortcuts list

- Ctrl+PgDn Switches between worksheet tabs, from left to right.
- Ctrl+PgUp Switches between worksheet tabs, from right to left.
- Ctrl+Shift+& Applies the outline border to the selected cells.
- Ctrl+Shift_ Removes the outline border from the selected cells.
- Ctrl+Shift+~ Applies the General number format.
- Ctrl+Shift+$ Applies the Currency format with two decimal places (negative numbers in parentheses).
- Ctrl+Shift+% Applies the Percentage format with no decimal places.
- Ctrl+Shift+^ Applies the Scientific number format with two decimal places.
- Ctrl+Shift+# Applies the Date format with the day, month, and year.
- Ctrl+Shift+@ Applies the Time format with the hour and minute, and AM or PM.

- Ctrl+Shift+! Applies the Number format with two decimal places, thousands separator, and minus sign (-) for negative values.
- Ctrl+Shift+* Selects the current region around the active cell (the data area enclosed by blank rows and blank columns). In a PivotTable, it selects the entire PivotTable report.
- Ctrl+Shift+: Enters the current time.
- Ctrl+Shift+" Copies the value from the cell above the active cell into the cell or the Formula Bar.
- Ctrl+Shift+Plus (+) Displays the Insert dialog box to insert blank cells.
- Ctrl+Minus (-) Displays the Delete dialog box to delete the selected cells.
- Ctrl+; Enters the current date.
- Ctrl+` Alternates between displaying cell values and displaying formulas in the worksheet.
- Ctrl+' Copies a formula from the cell above the active cell into the cell or the Formula Bar.
- Ctrl+1 Displays the Format Cells dialog box.
- Ctrl+2 Applies or removes bold formatting.
- Ctrl+3 Applies or removes italic formatting.
- Ctrl+4 Applies or removes underlining.
- Ctrl+5 Applies or removes strikethrough.
- Ctrl+6 Alternates between hiding and displaying objects.
- Ctrl+8 Displays or hides the outline symbols.
- Ctrl+9 Hides the selected rows.
- Ctrl+0 Hides the selected columns.
- Ctrl+A Selects the entire worksheet.
- If the worksheet contains data, Ctrl+A selects the current region. Pressing Ctrl+A a second time selects the entire worksheet. When the insertion point is to the right of a function name in a formula, it displays the Function Arguments dialog box.
- Ctrl+Shift+A inserts the argument names and parentheses when the insertion point is to the right of a function name in a formula.
- Ctrl+B Applies or removes bold formatting.
- Ctrl+C Copies the selected cells.
- Ctrl+D Uses the Fill Down command to copy the contents and format of the topmost cell of a selected range into the cells below.

- Ctrl+E Adds more values to the active column by using data surrounding that column.
- Ctrl+F Displays the Find and Replace dialog box, with the Find tab selected.
- Shift+F5 also displays this tab, while Shift+F4 repeats the last Find action.
- Ctrl+Shift+F opens the Format Cells dialog box with the Font tab selected.
- Ctrl+G Displays the Go To dialog box. F5 also displays this dialog box.
- Ctrl+H Displays the Find and Replace dialog box, with the Replace tab selected.
- Ctrl+I Applies or removes italic formatting.
- Ctrl+K Displays the Insert Hyperlink dialog box for new hyperlinks or the Edit Hyperlink dialog box for selected existing hyperlinks.
- Ctrl+L Displays the Create Table dialog box.
- Ctrl+N Creates a new, blank workbook.
- Ctrl+O Displays the Open dialog box to open or find a file.
- Ctrl+Shift+O selects all cells that contain comments.
- Ctrl+P Displays the Print tab in Microsoft Office Backstage view.
- Ctrl+Shift+P opens the Format Cells dialog box with the Font tab selected.
- Ctrl+Q Displays the Quick Analysis options for your data when you have cells that contain that data selected.
- Ctrl+R Uses the Fill Right command to copy the contents and format of the leftmost cell of a selected range into the cells to the right.
- Ctrl+S Saves the active file with its current file name, location, and file format.
- Ctrl+T Displays the Create Table dialog box.
- Ctrl+U Applies or removes underlining.
- Ctrl+Shift+U switches between expanding and collapsing of the formula bar.
- Ctrl+V Inserts the contents of the Clipboard at the insertion point and replaces any selection. Available only after you have cut or copied an object, text, or cell content.
- Ctrl+Alt+V displays the Paste Special dialog box. Available only after you have cut or copied an object, text, or cell content on a worksheet or in another program.
- Ctrl+W Closes the selected workbook window.
- Ctrl+X Cuts the selected cells.
- Ctrl+Y Repeats the last command or action, if possible.
- Ctrl+Z Uses the Undo command to reverse the last command or to delete the last entry that you typed.

Lastly, there is a last section in the Record Macro pop-up window, which focuses on where to store our macro.



There are three different options for storing macros in Excel:

1. This Workbook: This option stores the macro in the current workbook only. This means that the macro will only be available when that workbook is open.
2. Personal Macro Workbook: This is a hidden workbook that is created automatically by Excel when the user records their first macro. It is saved in the user's 'AppData' folder, and it can be accessed from any workbook in Excel. Any macros recorded in this workbook will be available whenever Excel is opened.
3. New Workbook: This option allows you to create a new workbook specifically for storing macros. The macro will be saved in this workbook only, and it can be accessed whenever the workbook is open.

The choice will depend on the intended use of the macro. If the macro is needed frequently across multiple workbooks, the 'Personal Macro Workbook' option may be the best choice. If the macro is specific to a particular workbook, the ‘This Workbook’ option may be more appropriate. If the user wants to create a separate workbook just for storing macros, they can choose the ‘New Workbook option’.

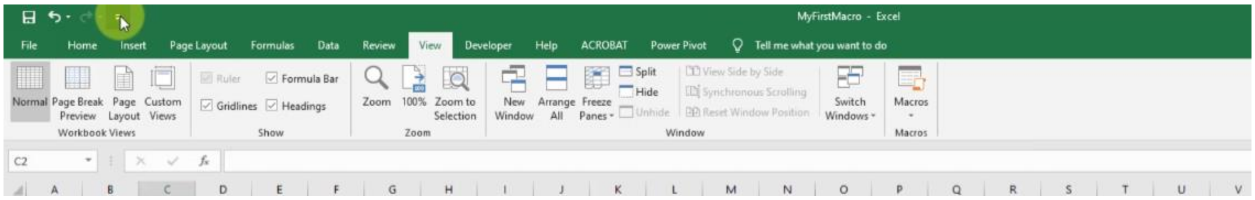
It's worth noting that macros can also be saved in an external file, such as a ‘.bas’ file. This file can be imported into Excel and executed as needed, but we will cover this feature later in the course.

Once you have recorded a macro, you can play back the macro at any time, and Excel will automatically execute each step in the sequence.

You can run the macros you record by clicking on the Developer tab and selecting the Macros button. A pop-up window will appear, and you can select the macro you want to run from the list.

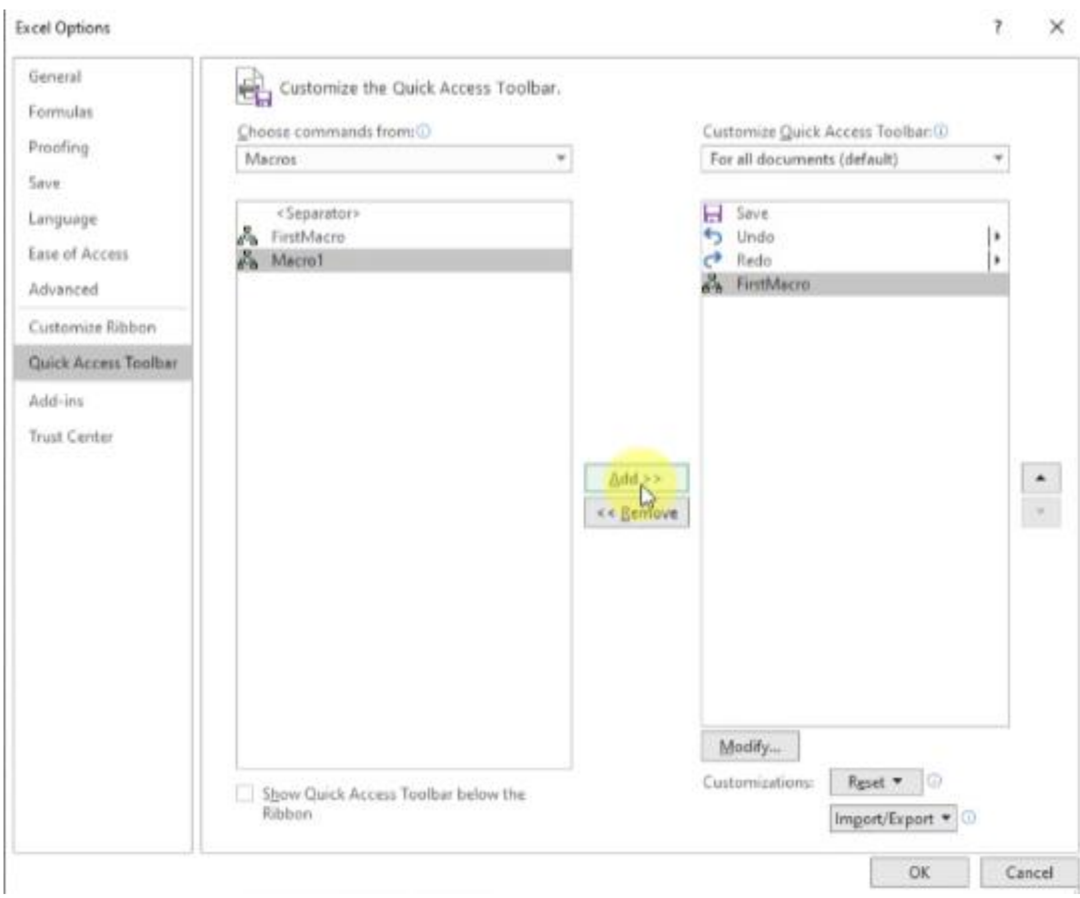
Advanced macro execution options

You can associate a macro to a button in the Quick Access toolbar of Excel.



Choose more commands

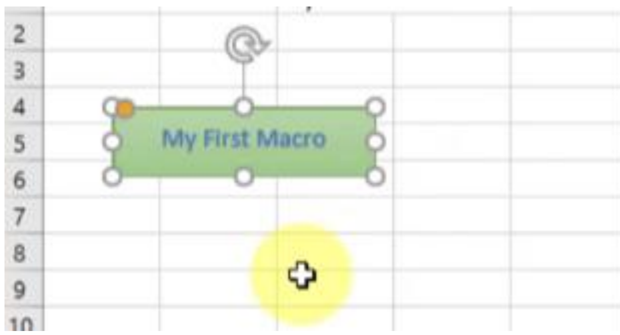
On your left side, from 'Choose commands from' click the dropdown arrow and then select the 'macros' option. Select the macro you want, and by clicking 'Add', you can add it to the quick access toolbar.

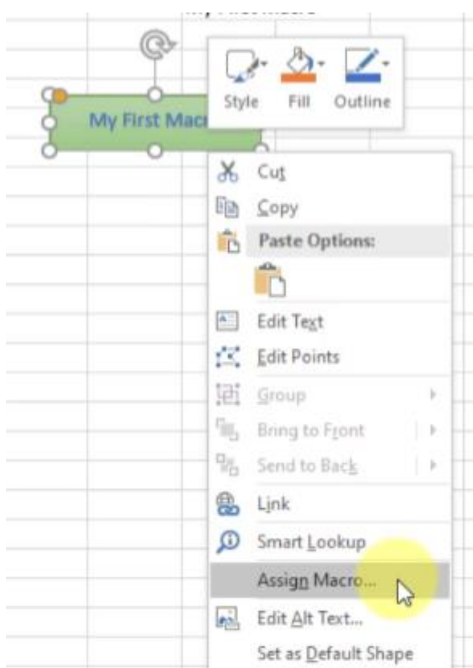


If you want to change that button, click Modify and you can choose the button you want from 200 different types

Another option to run the macro is assigning it to an Excel element, such as a shape, a picture, or a form control. You cannot expect another user of your macro to run it using the Developer tab.

First, create the form you want the user to interact with to run the macro. Then, right-click and select assign Macro.

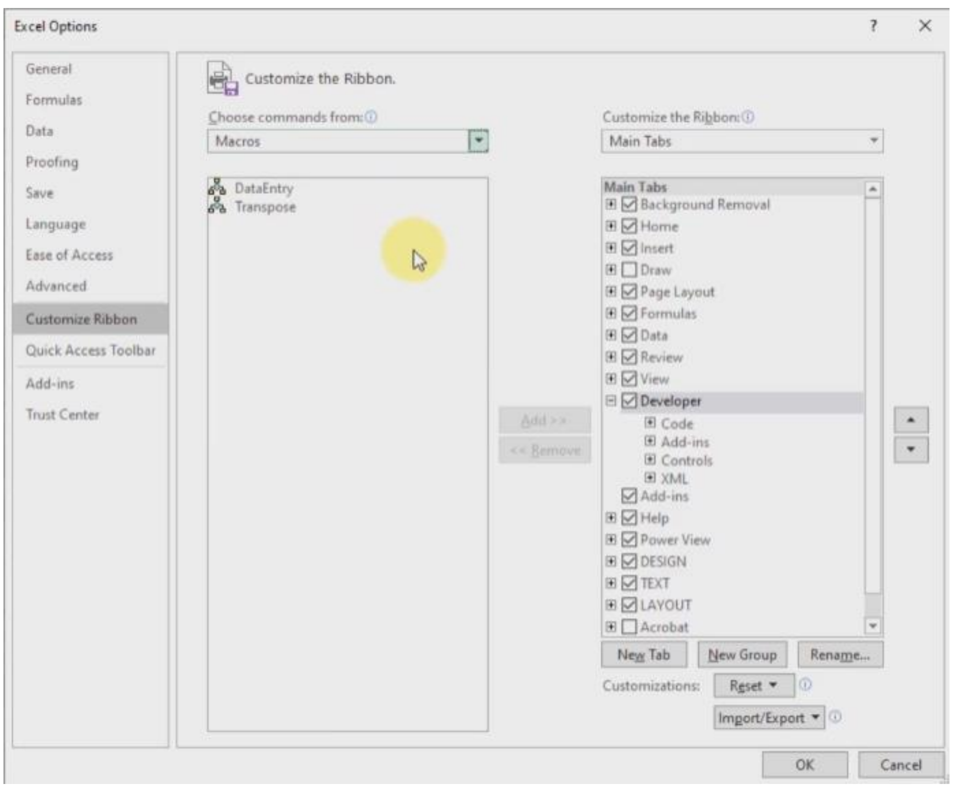




We've seen how to add buttons and shapes to trigger a macro, which is great, but they are only relevant to macros that work in a single sheet. Moreover, if we add a button to the quick access toolbar, it can become messy.

Therefore, the last option to add a macro visual element in Excel is to add a macro to the Excel tab!

1. Go to the File tab.
2. Click on the options button.
3. Click on “customized ribbon”.
4. Click the dropdown and choose macros.



With this option, you can add a macro to any existing tab of Excel, or even create a new tab that contains your macros!

Visual Basic editor

Once we stop recording our macro, Excel stores it in a new module automatically and names it Module1. But... where can you see the result of the recording?

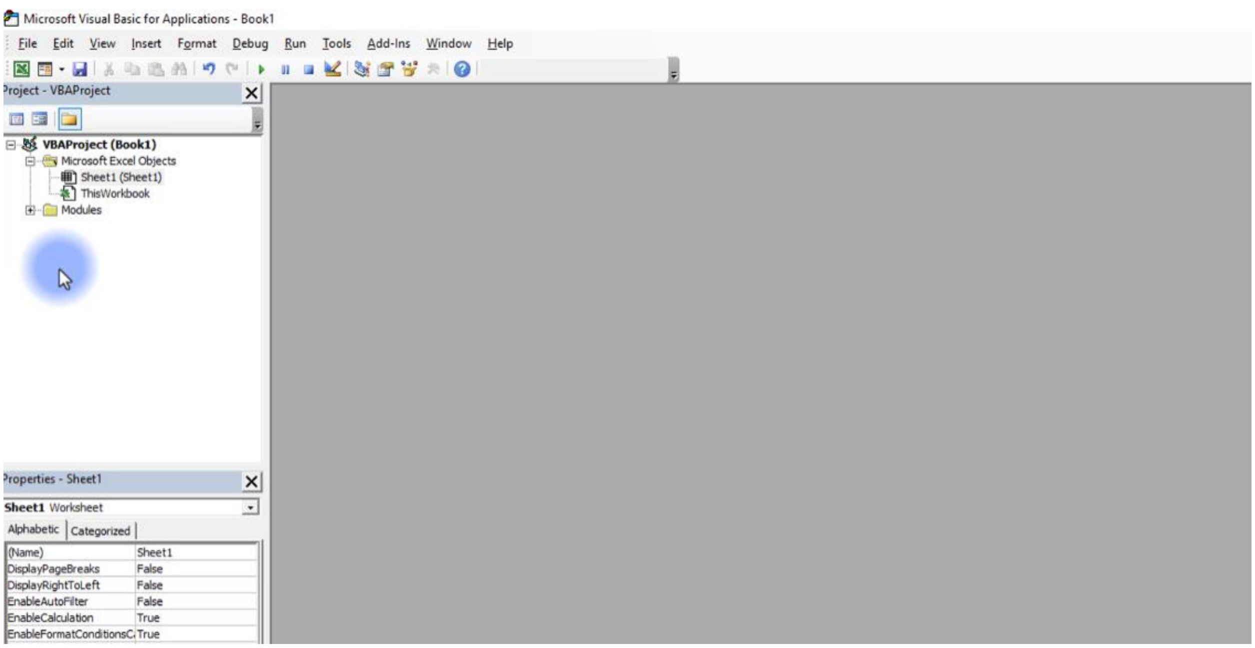
Well, that's why you use the Visual Basic Editor (VBE). It is a tool that allows you to create, edit, and manage Visual Basic for Applications (VBA) code modules. It is a development environment that is integrated with Excel and provides several features that make it easy to write and test VBA code.

The VBE can be accessed by clicking on the Developer tab in the Ribbon, and then selecting 'Visual Basic' from the Code group.

Another way to open the VBE is by using the following shortcuts:

- Alt + F11 (Windows)
- Fn + Opt + F11 (macOS)

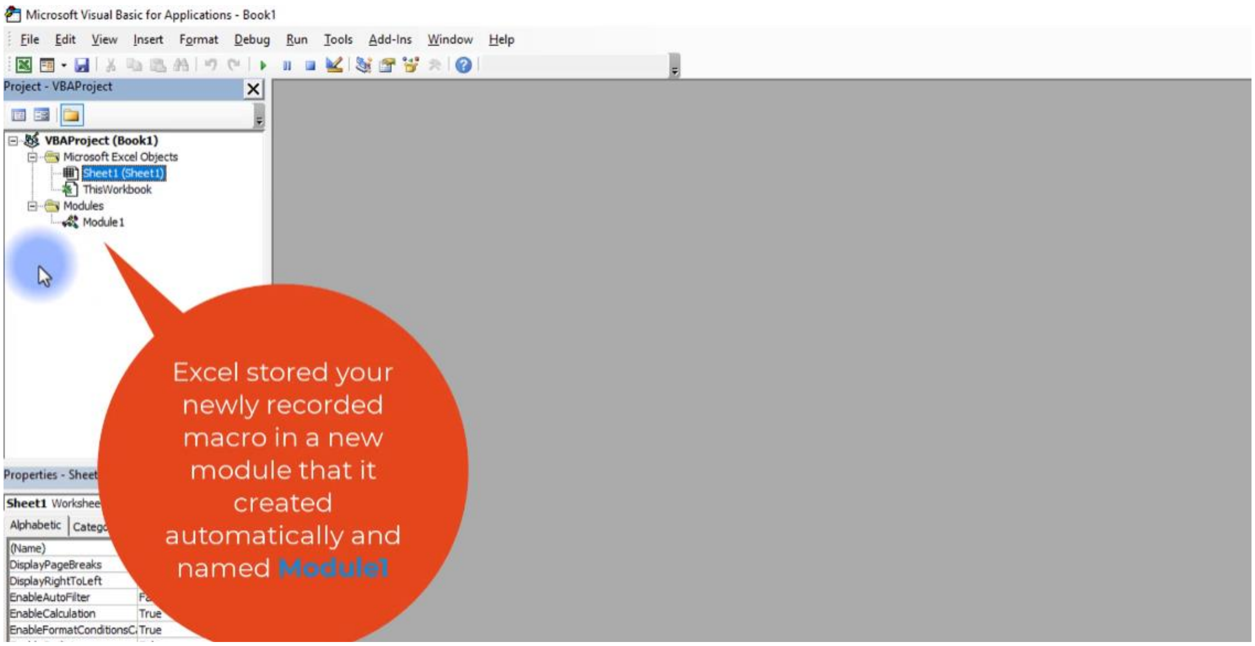
This will open the VBE window, which consists of different panes and windows that allow users to work with VBA code.



The VBE provides many features:

- Code window: This is where users can write, check, and edit VBA code.
- Project Explorer: This is where users can manage VBA projects and modules.
- Properties window: This is where users can view and edit the properties of objects in their VBA code.

The code that you recorded previously is stored in module one in the current workbook. When you double-click module one, the code in the module appears in the code window.



Remember the 'personal macro workbook' save option? You can see the macros stored on it inside the VBE! (See page. 15)

You can open Module1 if you want, but I think at this moment you won't be able to 'read' and understand the VBA code that is inside. So, let's continue our journey!